

```
In [1]: #Step 5:
        %cd size400
        !python ../kbencoder.py --generate_kb --num_rules 400 --new_vocab --save_vocab
```

```
/data/users/jupyter-rem526/cse327-p4-files/size400
Classes: 1000000000.0
Max arity: 4
KB generated (400)
```

```
In [3]: %cd size400
```

```
/data/users/jupyter-rem526/cse327-p4-files/size400
```

```
In [4]: #Step 6, train random KB of size 400
        !python ../kbencoder.py --kb_path randomKB.txt --train_unification_model
```

```
Using cuda device
```

```
Uni model embed size: 50
```

```
[██████████] 3500/3500 Anchors generated
[██████████] 70000/70000 Triplets generated
[██████-----] 18410/70000 Triplet encodings
```

IOPub data rate exceeded.
 The Jupyter server will temporarily stop sending output
 to the client in order to avoid crashing it.
 To change this limit, set the config variable
 `--ServerApp.iopub\_data\_rate\_limit`.

```
Current values:
```

```
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
```

```
ServerApp.rate_limit_window=3.0 (secs)
```

```
[██████████-----] 47535/70000 Triplet encodings
```

IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

[████████████████████] 70000/70000 Triplet encodings
Early stopping triggered-----] 19.8% Epoch | [Epoch 100/500, Training Loss: 0.2345, Validation Loss: 0.4849]
Saved model.
Saved plot.
unification model generated

In [5]: !python ../kbencoder.py --kb_path randomKB.txt -g
!python ../kbencoder.py --kb_path randomKB.txt -p

Generating facts list...
Forward chaining depth: 1
New facts: 173
Forward chaining depth: 2
New facts: 61
Forward chaining depth: 3
New facts: 12
Forward chaining depth: 4
New facts: 1
Forward chaining depth: 5

Facts list generated
247 total facts. Max depth = 4

Number of queries: 200
100 training queries generated
100 testing queries generated
running training queries...
Negative facts will be generated
1: [p4(a66, X2)]
min depth: 2
Nodes: 4487
Answers: 2

2: [p8(X2, a27, a22, X0), p6(X0, X1), p2(a11, X1), p5(a57, a1, X2)]
min depth: 1
Nodes: 2
Answers: 1

3: [p0(X1, X0), p7(X0, a24), p0(X1, a11), p8(a11, X0, a76, a12), p2(a75, X1)]
min depth: 1
Nodes: 20247
Answers: 93

4: [p2(X6, X3)]
min depth: 1
Nodes: 24253
Answers: 146

5: [p4(X4, X2)]
min depth: 1
Nodes: 29120

Answers: 200

6: [p5(X3, a0, a0)]

min depth: 2

Nodes: 30285

Answers: 1

7: [p2(X3, a73)]

Restart...

min depth: 2

Nodes: 3060

Answers: 3

8: [p4(X0, a57)]

Restart...

min depth: 3

Nodes: 39146

Answers: 140

9: [p5(a57, X8, X3)]

min depth: 2

Nodes: 409

Answers: 97

10: [p8(X2, a91, X0, X1), p5(X0, a53, a66), p1(X1, a0), p5(a57, a32, X1), p2(a10, X2)]

min depth: 1

Nodes: 2

Answers: 1

11: [p8(a30, a27, X7, X5)]

min depth: 2

Nodes: 566

Answers: 4

12: [p4(X0, X6)]

min depth: 1

Nodes: 10953

Answers: 150

13: [p0(X5, X8)]

min depth: 1

Nodes: 8377

Answers: 46

14: [p0(X0, X0)]

Restart...

min depth: 2

Nodes: 8111

Answers: 3

15: [p2(X5, X3)]

min depth: 1

Nodes: 100059

Answers: 147

16: [p4(a82, X7)]

Restart...

Restart...

Answers: 0

17: [p4(X1, X7)]

min depth: 1

Nodes: 15925

Answers: 149

18: [p1(X0, a0), p0(X0, X1), p6(X1, a51)]

min depth: 3

Nodes: 336

Answers: 6

19: [p2(X1, X0), p2(a10, X0), p2(a38, X0), p6(X1, a10)]

min depth: 1

Nodes: 21497

Answers: 139

20: [p8(a0, X0, a31, a0), p5(X1, X0, a10), p5(X1, a76, a79)]

min depth: 3

Nodes: 276

Answers: 2

21: [p2(a75, X0), p2(X1, X0), p1(a84, X1)]

Restart...

min depth: 1

Nodes: 31373

Answers: 2

22: [p5(a14, a93, X0), p4(X0, X0)]

min depth: 1

Nodes: 48

Answers: 2

23: [p8(a20, a63, X0, a5), p6(X0, a70)]

min depth: 1

Nodes: 2

Answers: 1

24: [p4(X2, X6)]

min depth: 1

Nodes: 15688

Answers: 345

25: [p3(X7, a20)]

min depth: 1

Nodes: 34

Answers: 30

26: [p5(a0, X9, X8)]

min depth: 1

Nodes: 26558

Answers: 19

27: [p2(X5, a0)]

Nodes: 100102

Answers: 6

28: [p5(X1, X4, X9)]

min depth: 1

Nodes: 52035

Answers: 39

29: [p4(X6, X3)]

min depth: 1

Nodes: 6552

Answers: 174

30: [p4(X0, X1), p2(a79, X0), p5(X1, X0, a21), p5(X1, a25, a43)]

min depth: 1
[-----] 12.0% to max nodes (4285 nps)Mismatched predicates: p2(a79, X0) with p8(X00, X0, a71, X10)
Nodes: 91579
Answers: 491

31: [p5(X0, a1, X9)]
Nodes: 100053
Answers: 1

32: [p4(a53, X0)]
Restart...
min depth: 2
Nodes: 94
Answers: 1

33: [p2(X1, X0), p1(X1, X0), p5(a49, X2, X1), p8(X0, X2, a79, a52), p2(X2, a12)]
min depth: 1
Nodes: 58288
Answers: 174

34: [p4(a66, X0), p8(X0, a27, a50, a22)]
min depth: 2
Nodes: 1040
Answers: 3

35: [p2(X1, X1)]
Restart...
Restart...
Answers: 0

36: [p4(a68, X0), p4(a91, X0)]
min depth: 4
Mismatched predicates: p4(a91, X0) with p8(X00, a68, a71, X10)
Nodes: 6562
Answers: 58

37: [p4(X0, a57), p2(a9, X0), p8(X0, a11, a29, a39)]
[-----] 0.0% to max nodes (58626 nps)Mismatched predicates: p2(a9, X0) with p8(X00, a57, a71, X10)
Restart...
[-----] 0.0% to max nodes (84925 nps)Mismatched predicates: p2(a9, X0) with p8(X00, a57, a71, X10)
Restart...
Answers: 0

38: [p2(X7, X5)]
min depth: 1
Nodes: 78706
Answers: 73

39: [p2(X2, a21)]
Restart...
Restart...
min depth: 2
Nodes: 11752
Answers: 1

40: [p4(a82, X4)]
Restart...
Restart...
Answers: 0

41: [p0(X9, a55)]
Restart...
min depth: 3
Nodes: 23441
Answers: 2

42: [p5(a57, a56, X0), p2(X0, a0), p2(a0, X0), p8(a11, a27, a50, X0)]
Restart...
Restart...
Nodes: 100159
Answers: 2

43: [p0(a17, X0)]
Restart...
Restart...
min depth: 2
Nodes: 650
Answers: 29

44: [p2(X1, X8)]
min depth: 1
Nodes: 25625
Answers: 113

45: [p5(a65, X0, a95), p5(X0, a13, a55), p4(X0, a35)]
min depth: 1
Nodes: 714
Answers: 1

46: [p9(X0), p5(a0, a57, X0), p7(X0, a0)]
min depth: 1
Nodes: 30
Answers: 29

47: [p2(X7, a0)]
min depth: 1
Nodes: 924
Answers: 26

48: [p4(a45, X2)]
Nodes: 100025
Answers: 1

49: [p5(X6, X5, X5)]
Restart...
Restart...
Answers: 0

50: [p1(X0, X1), p0(X0, a84), p1(X1, a1), p8(X0, a47, a82, a1)]
min depth: 1
Nodes: 3294
Answers: 44

51: [p8(a23, a27, X0, a22), p2(a28, X0), p8(X0, a11, a87, a62)]
Nodes: 100084
Answers: 1

52: [p2(X0, X1), p7(X0, a35), p0(X1, X0), p9(X1)]
min depth: 1
Nodes: 66917
Answers: 106

53: [p8(X7, X8, a50, X9)]
Nodes: 100130
Answers: 36

54: [p4(a6, a0)]

min depth: 2

Nodes: 1467

Answers: 3

55: [p2(a32, X5)]

Restart...

Restart...

Answers: 0

56: [p8(X0, a63, X1, a0), p9(X0), p4(X1, X2), p8(X3, a27, a50, X2), p1(X3, a41)]

min depth: 1

Nodes: 2

Answers: 1

57: [p5(a66, a48, X0), p2(a52, X0)]

Restart...

Restart...

min depth: 1

Nodes: 485

Answers: 1

58: [p4(X1, a29)]

Nodes: 100107

Answers: 2

59: [p4(a20, X1)]

Nodes: 100109

Answers: 1

60: [p4(X5, a24)]

min depth: 1

Nodes: 2767

Answers: 2

61: [p5(a57, a28, X3)]

Restart...

min depth: 2

Nodes: 10774

Answers: 2

62: [p2(X7, X5)]

min depth: 1
Nodes: 94555
Answers: 172

63: [p1(X1, X0), p3(X0, X2), p4(a10, X1), p2(a27, X2)]
min depth: 1
Nodes: 8766
Answers: 43

64: [p2(X3, X9)]
min depth: 1
Nodes: 6333
Answers: 149

65: [p4(X1, X0), p6(a13, X0), p4(X1, a16)]
min depth: 1
[-----] 2.7% to max nodes (4677 nps)Mismatched predicates: p6(a13, X0) with p8(X00, X1, a71, X10)
Nodes: 27651
Answers: 264

66: [p4(X4, X3)]
min depth: 1
Nodes: 48884
Answers: 269

67: [p4(X8, X0)]
min depth: 1
Nodes: 20081
Answers: 282

68: [p4(a19, X0), p0(a0, X0)]
min depth: 1
Mismatched predicates: p0(a0, X0) with p8(X00, a19, a71, X10)
Nodes: 15918
Answers: 16

69: [p3(X0, a73), p3(X0, a10)]
min depth: 1
Mismatched predicates: p3(X0, a10) with p6(a49, a73)
Nodes: 5
Answers: 2

70: [p5(X8, X0, a2)]

Nodes: 100117

Answers: 2

71: [p4(a1, X7)]

min depth: 1

Nodes: 1198

Answers: 10

72: [p5(X8, X7, X2)]

min depth: 1

Nodes: 13129

Answers: 39

73: [p5(a57, a19, a55)]

min depth: 2

Nodes: 8799

Answers: 1

74: [p7(X0, a5), p1(X0, X1), p6(a54, X1), p8(a84, X0, X2, a22), p8(a0, X0, X2, a22)]

min depth: 1

Nodes: 66

Answers: 2

75: [p4(a25, X0), p2(X0, X1), p5(X2, X1, a94), p4(a24, X2)]

Nodes: 100197

Answers: 2

76: [p2(a30, a51)]

min depth: 2

Nodes: 69

Answers: 1

77: [p4(a82, X7)]

Restart...

Restart...

min depth: 5

Nodes: 54782

Answers: 93

78: [p4(X5, X2)]

min depth: 1

Nodes: 41330

Answers: 255

79: [p6(a26, X0), p8(a85, X0, a21, a10), p5(X0, X1, a92), p4(a10, X1), p3(X1, a94)]

min depth: 1

Nodes: 3

Answers: 2

80: [p2(X5, X7)]

min depth: 1

Nodes: 40471

Answers: 73

81: [p3(X9, a20)]

min depth: 2

Nodes: 36

Answers: 29

82: [p4(X3, X9)]

min depth: 1

Nodes: 19251

Answers: 152

83: [p3(a40, X0), p6(a59, X0), p6(a46, X0)]

Nodes: 100189

Answers: 1

84: [p7(a10, X0), p1(a84, X0)]

min depth: 1

Nodes: 201

Answers: 3

85: [p1(a0, X0), p2(X1, X0), p0(X0, X1), p3(a62, X0)]

min depth: 1

Nodes: 8

Answers: 7

86: [p8(a77, a27, a50, X2)]

min depth: 2

Nodes: 1211

Answers: 2

87: [p2(a30, X5)]
min depth: 2
Nodes: 1415
Answers: 5

88: [p2(X1, X0)]
min depth: 1
Nodes: 48109
Answers: 127

89: [p5(X1, X0, a95), p7(X0, a81), p8(a0, a27, X1, a81)]
Nodes: 100091
Answers: 1

90: [p5(a57, X8, X5)]
min depth: 2
Nodes: 20709
Answers: 193

91: [p4(X0, a16), p0(X0, X1), p1(X1, a4)]
min depth: 2
Mismatched predicates: p0(X0, X1) with p8(X00, a16, a71, X10)
Nodes: 9947
Answers: 16

92: [p2(X4, X5)]
min depth: 1
Nodes: 11803
Answers: 192

93: [p8(X6, a18, a31, X6)]
Restart...
min depth: 3
Nodes: 15023
Answers: 9

94: [p4(X9, a10)]
min depth: 1
Nodes: 10926
Answers: 7

95: [p4(X2, a16)]

min depth: 1
Nodes: 4688
Answers: 11

96: [p8(a25, a66, X0, a36), p8(a39, a0, X1, X0), p4(X1, a85)]
min depth: 1
Nodes: 2
Answers: 1

97: [p0(X2, a1)]
min depth: 1
Nodes: 1029
Answers: 6

98: [p2(a9, X0), p2(X0, X1), p4(X1, X2), p2(X1, a0), p8(a16, X2, a81, X2)]
Nodes: 100103
Answers: 1

99: [p2(X6, X7)]
min depth: 1
Nodes: 16231
Answers: 177

100: [p5(X0, a82, a2), p6(a0, X0), p2(a11, X0)]
Restart...
Restart...
Answers: 0

Before removing duplicates: 3012181 examples
After removing duplicates: 29321 examples
Replicating pos examples 2 times
pos len=22509, neg len=22509

In [6]: !python ../nnreasoner.py --embed_type unification -s randomKB_400_model.pt

```

States from vocab: vocab
Num predicates: 10
Max arity: 4
Num constants: 100
Num variables: 10

Training unification
Embed size: 50
Reading examples from mr_train_examples.csv
Parsing examples...
Generating embeddings...
Embedding input size: 450
[██████████] 45017/45017 Prepping examples
Loaded data...
Using cuda device
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
0      0.23895 (-)
50     0.06183 (-0.0016)
100    0.05753 (-0.0008)
150    0.05563 (-0.0006)
200    0.05445 (-0.0004)
250    0.05338 (-0.0003)
300    0.05264 (-0.0003)
350    0.05195 (-0.0002)
400    0.05133 (-0.0002)
450    0.05091 (-0.0002)
500    0.05051 (-0.0001)
550    0.04994 (-0.0001)
600    0.04940 (-0.0001)
650    0.04932 (-0.0001)
700    0.04903 (-0.0001)
749    0.04867 (-9.0661)
Saved training loss figure.

```

```
In [20]: !python ../evaluate.py --kb randomKB.txt -s
```


using cuda device
STANDARD

Query 1: [p4(X5, a14)]
4 :: 2 - 00:00:00 (2851 nps)

Query 2: [p4(X3, a57)]
5 :: 3 - 00:00:00 (2736 nps)

Query 3: [p9(X0), p3(X0, a12)]
Query failed!!!
1000000 :: 0 - 00:01:14 (13451 nps)

Query 4: [p2(X0, a28), p4(a21, X0)]
3 :: 2 - 00:00:00 (3085 nps)

Query 5: [p7(X0, X1), p7(X0, a1), p6(a2, X1), p6(X1, X2), p2(X2, a99)]
297 :: 5 - 00:00:00 (6687 nps)

Query 6: [p4(X3, X0)]
2 :: 1 - 00:00:00 (3504 nps)

Query 7: [p4(a1, X4)]
2 :: 1 - 00:00:00 (4117 nps)

Query 8: [p8(a40, a27, X5, X0)]
3 :: 2 - 00:00:00 (2987 nps)

Query 9: [p2(X0, a99), p7(X0, a16), p6(a39, X0)]
4 :: 3 - 00:00:00 (3968 nps)

Query 10: [p2(X1, X7)]
2 :: 1 - 00:00:00 (3455 nps)

Query 11: [p5(a82, a12, X0), p7(X1, X0), p6(a36, X1), p1(a36, X1)]
5 :: 4 - 00:00:00 (3804 nps)

Query 12: [p8(a41, a27, a50, X9)]
3 :: 2 - 00:00:00 (3609 nps)

Query 13: [p2(X2, X8)]
2 :: 1 - 00:00:00 (3254 nps)

Query 14: [p8(a71, a27, X5, a22)]
3 :: 2 - 00:00:00 (3881 nps)

Query 15: [p4(a82, X0)]
Query failed!!!
1000000 :: 0 - 00:01:05 (15365 nps)

Query 16: [p2(X4, X0)]
2 :: 1 - 00:00:00 (2709 nps)

Query 17: [p5(a57, a20, X7)]
3 :: 2 - 00:00:00 (2974 nps)

Query 18: [p2(X1, X0), p4(X0, a57), p4(X1, a57), p4(X0, a10), p2(a79, X1)]
Query failed!!!
1000000 :: 0 - 00:00:59 (16823 nps)

Query 19: [p4(a11, a57)]
4 :: 3 - 00:00:00 (2459 nps)

Query 20: [p3(X0, a95), p4(X1, X0), p3(X1, a27), p4(X2, X1), p4(X2, a5)]
31 :: 6 - 00:00:00 (2857 nps)

Query 21: [p5(a15, X6, X3)]
2 :: 1 - 00:00:00 (3565 nps)

Query 22: [p5(a57, a8, a1)]
3 :: 2 - 00:00:00 (3783 nps)

Query 23: [p4(a47, X0)]
5 :: 3 - 00:00:00 (2555 nps)

Query 24: [p4(X4, X5)]
2 :: 1 - 00:00:00 (3196 nps)

Query 25: [p2(X0, a17), p2(a79, X0)]
Query failed!!!
1000000 :: 0 - 00:01:02 (15939 nps)

Query 26: [p7(X2, X0), p2(X0, X1), p2(X3, X1), p4(a36, X2), p5(a57, X3, a79)]
Query failed!!!

1000000 :: 0 - 00:01:18 (12764 nps)

Query 27: [p2(a12, X3)]

Query failed!!!

1000000 :: 0 - 00:01:00 (16559 nps)

Query 28: [p4(X1, X0), p2(X0, X1), p7(a23, X1), p1(X0, X0), p8(X0, X1, a38, a49)]

Query failed!!!

1000000 :: 0 - 00:01:32 (10774 nps)

Query 29: [p2(X0, a78), p0(X0, X1), p2(a43, X1), p4(X1, a57)]

8 :: 7 - 00:00:00 (2521 nps)

Query 30: [p2(X1, X0), p8(X0, X2, a29, a11), p5(a0, a57, X1), p5(a57, X2, a92)]

Query failed!!!

1000000 :: 0 - 00:01:12 (13818 nps)

Query 31: [p4(a93, X0), p8(X2, a13, X0, X1), p4(a6, X1), p8(a17, X2, a16, X1)]

8 :: 6 - 00:00:00 (2289 nps)

Query 32: [p8(X2, a27, X8, a22)]

3 :: 2 - 00:00:00 (2627 nps)

Query 33: [p4(X4, X5)]

2 :: 1 - 00:00:00 (2957 nps)

Query 34: [p5(a57, a10, X0), p5(X2, X0, X1), p4(X0, X2), p5(a11, X0, X1), p2(a25, X2)]

Query failed!!!

1000000 :: 0 - 00:01:00 (16506 nps)

Query 35: [p5(X0, a40, X1), p5(X0, a62, a73), p5(X2, a38, X1), p4(a42, X0), p9(X2)]

10 :: 9 - 00:00:00 (2518 nps)

Query 36: [p5(X2, X6, a17)]

3 :: 2 - 00:00:00 (3415 nps)

Query 37: [p2(X2, X6)]

2 :: 1 - 00:00:00 (2737 nps)

Query 38: [p4(a10, X1)]

2 :: 1 - 00:00:00 (3669 nps)

Query 39: [p2(X0, a92), p8(X0, a27, a50, a22)]

Query failed!!!

1000000 :: 0 - 00:01:04 (15573 nps)

Query 40: [p4(X1, X0), p4(a22, X0), p3(X1, a20)]

Query failed!!!

1000000 :: 0 - 00:01:03 (15767 nps)

Query 41: [p8(a26, a11, X0, a39), p1(X0, X1), p5(a57, X1, a2)]

5 :: 4 - 00:00:00 (2622 nps)

Query 42: [p8(X8, X2, a50, X3)]

2 :: 1 - 00:00:00 (3363 nps)

Query 43: [p5(X7, X1, a11)]

3 :: 2 - 00:00:00 (3630 nps)

Query 44: [p0(X4, X4)]

4 :: 2 - 00:00:00 (2950 nps)

Query 45: [p4(a42, X0)]

5 :: 3 - 00:00:00 (2563 nps)

Query 46: [p4(X7, X6)]

2 :: 1 - 00:00:00 (3160 nps)

Query 47: [p1(X0, a37), p4(a10, X0)]

3 :: 2 - 00:00:00 (3741 nps)

Query 48: [p5(X8, a68, X2)]

Query failed!!!

1000000 :: 0 - 00:01:16 (13011 nps)

Query 49: [p5(X6, a2, X0)]

3 :: 2 - 00:00:00 (2675 nps)

Query 50: [p2(a11, X3)]

Query failed!!!

1000000 :: 0 - 00:01:18 (12755 nps)

Query 51: [p2(X1, X7)]

2 :: 1 - 00:00:00 (2659 nps)

Query 52: [p8(a55, X8, a50, X4)]
8 :: 4 - 00:00:00 (2838 nps)

Query 53: [p2(a42, X3)]
Query failed!!!
1000000 :: 0 - 00:01:17 (12920 nps)

Query 54: [p2(X8, a51)]
Query failed!!!
1000000 :: 0 - 00:01:12 (13840 nps)

Query 55: [p4(X5, a57)]
5 :: 3 - 00:00:00 (2153 nps)

Query 56: [p5(a57, a26, X0), p0(a16, X0)]
4 :: 3 - 00:00:00 (2880 nps)

Query 57: [p8(X0, a27, X6, X3)]
2 :: 1 - 00:00:00 (2990 nps)

Query 58: [p2(a32, a10)]
3 :: 2 - 00:00:00 (3358 nps)

Query 59: [p2(X3, a92)]
Query failed!!!
1000000 :: 0 - 00:00:58 (17052 nps)

Query 60: [p0(a14, X3)]
4 :: 3 - 00:00:00 (2708 nps)

Query 61: [p9(X0), p1(a0, X0)]
4 :: 2 - 00:00:00 (3519 nps)

Query 62: [p2(X4, X5)]
2 :: 1 - 00:00:00 (2729 nps)

Query 63: [p4(X3, X4)]
2 :: 1 - 00:00:00 (3055 nps)

Query 64: [p2(X0, X1), p0(X1, X0), p2(X1, a79), p1(a2, X0)]
Query failed!!!

1000000 :: 0 - 00:01:08 (14593 nps)

Query 65: [p4(a13, X5)]

4 :: 2 - 00:00:00 (2301 nps)

Query 66: [p2(X1, a14)]

2 :: 1 - 00:00:00 (3682 nps)

Query 67: [p2(X0, X1), p0(X1, X0), p5(a57, X1, X2), p1(X1, X2), p3(X1, X2)]

Query failed!!!

1000000 :: 0 - 00:01:09 (14346 nps)

Query 68: [p3(X0, X0), p5(a57, X0, X0)]

4 :: 3 - 00:00:00 (2774 nps)

Query 69: [p8(a30, a27, X1, a22)]

3 :: 2 - 00:00:00 (3203 nps)

Query 70: [p5(a57, a50, a0)]

3 :: 2 - 00:00:00 (3720 nps)

Query 71: [p8(a85, a18, X0, a43), p8(a5, a16, a42, X0)]

3 :: 2 - 00:00:00 (3421 nps)

Query 72: [p2(a4, X0), p4(a82, X0)]

8 :: 6 - 00:00:00 (2613 nps)

Query 73: [p2(X4, a0)]

2 :: 1 - 00:00:00 (3985 nps)

Query 74: [p2(X6, X5)]

2 :: 1 - 00:00:00 (3032 nps)

Query 75: [p1(a0, X0), p1(X0, X1), p2(X1, X2), p2(a36, X2)]

Query failed!!!

1000000 :: 0 - 00:01:13 (13553 nps)

Query 76: [p4(X0, a90), p3(X0, X1), p2(X1, a1)]

38046 :: 5 - 00:00:02 (18479 nps)

Query 77: [p4(X0, a5), p5(a1, X0, a64)]

18 :: 2 - 00:00:00 (4103 nps)

Query 78: [p5(X0, X2, X1), p5(X0, a2, X3), p3(X2, X1), p3(X2, a10), p4(X3, a90)]

Query failed!!!

1000000 :: 0 - 00:01:07 (14844 nps)

Query 79: [p3(X2, a20)]

2 :: 1 - 00:00:00 (3594 nps)

Query 80: [p4(X1, X2)]

2 :: 1 - 00:00:00 (2632 nps)

Query 81: [p8(X8, a27, a50, X3)]

3 :: 2 - 00:00:00 (2261 nps)

Query 82: [p5(a57, X0, a1), p4(a0, X0)]

4 :: 3 - 00:00:00 (2843 nps)

Query 83: [p4(X1, X0), p4(a20, X0), p5(X0, X1, a12), p2(a13, X1)]

281903 :: 9 - 00:00:22 (12806 nps)

Query 84: [p2(X8, a0)]

2 :: 1 - 00:00:00 (3129 nps)

Query 85: [p0(X1, X0), p8(X0, a63, a35, a27), p2(a8, X1)]

21 :: 3 - 00:00:00 (3027 nps)

Query 86: [p8(a53, X1, a50, X0), p8(X2, X1, a50, X0), p9(X1), p8(X2, a11, a29, a39)]

39 :: 6 - 00:00:00 (2501 nps)

Query 87: [p2(X3, X7)]

2 :: 1 - 00:00:00 (1301 nps)

Query 88: [p2(X2, a92)]

Query failed!!!

1000000 :: 0 - 00:00:59 (16880 nps)

Query 89: [p8(a26, a11, a29, X0), p6(a2, X0), p8(a20, a63, X0, a5)]

4 :: 3 - 00:00:00 (2794 nps)

Query 90: [p4(X4, X3)]

2 :: 1 - 00:00:00 (2927 nps)

Query 91: [p2(X0, a92), p6(X0, a10)]

Query failed!!!

1000000 :: 0 - 00:00:57 (17385 nps)

Query 92: [p5(a87, X0, a12), p1(X0, a1)]

3 :: 2 - 00:00:00 (3028 nps)

Query 93: [p4(a84, a52)]

3 :: 2 - 00:00:00 (3149 nps)

Query 94: [p8(X2, X0, X1, a22), p8(a10, X0, X1, a22), p8(a92, X0, X1, a22), p2(a1, X2), p2(X1, a92)]

Query failed!!!

1000000 :: 0 - 00:01:16 (13128 nps)

Query 95: [p5(X0, a57, a23)]

Query failed!!!

1000000 :: 0 - 00:01:15 (13234 nps)

Query 96: [p2(X0, X1)]

2 :: 1 - 00:00:00 (2510 nps)

Query 97: [p8(X9, a18, a31, X9)]

4 :: 3 - 00:00:00 (2983 nps)

Query 98: [p8(a58, X0, a50, a22), p8(a26, X0, a50, a22)]

5 :: 4 - 00:00:00 (2924 nps)

Query 99: [p1(X0, a4), p8(a25, X1, a6, X0), p5(a15, X1, a14)]

5 :: 3 - 00:00:00 (3014 nps)

Query 100: [p1(a0, X0), p1(X0, a1)]

6 :: 2 - 00:00:00 (3070 nps)

std: 243205.95 nodes/query

24 queries failed

Time to run all queries: 1689.3718287820006

In [19]: !python ../evaluate.py --kb randomKB.txt -u --scoring_model_path randomKB_400_model.pt


```
using cuda device
UNITY: ming
    Embedding Model: rKB_model.pth
    Scoring Model: randomKB_400_model.pt

    Control: mingoal
    Rule Eval: Max Rule
Learned Embedding: Network using cuda
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
Query 1: [p4(X5, a14)]
3 :: 2 - 00:00:00 (7 nps)

Query 2: [p4(X3, a57)]
4 :: 3 - 00:00:00 (116 nps)

Query 3: [p9(X0), p3(X0, a12)]
4 :: 2 - 00:00:00 (177 nps)

Query 4: [p2(X0, a28), p4(a21, X0)]
3 :: 2 - 00:00:00 (327 nps)

Query 5: [p7(X0, X1), p7(X0, a1), p6(a2, X1), p6(X1, X2), p2(X2, a99)]
201 :: 5 - 00:00:00 (384 nps)

Query 6: [p4(X3, X0)]
2 :: 1 - 00:00:00 (52 nps)

Query 7: [p4(a1, X4)]
2 :: 1 - 00:00:00 (720 nps)

Query 8: [p8(a40, a27, X5, X0)]
3 :: 2 - 00:00:00 (475 nps)

Query 9: [p2(X0, a99), p7(X0, a16), p6(a39, X0)]
4 :: 3 - 00:00:00 (228 nps)

Query 10: [p2(X1, X7)]
2 :: 1 - 00:00:00 (65 nps)

Query 11: [p5(a82, a12, X0), p7(X1, X0), p6(a36, X1), p1(a36, X1)]
```

5 :: 4 - 00:00:00 (110 nps)

Query 12: [p8(a41, a27, a50, X9)]

3 :: 2 - 00:00:00 (427 nps)

Query 13: [p2(X2, X8)]

2 :: 1 - 00:00:00 (65 nps)

Query 14: [p8(a71, a27, X5, a22)]

3 :: 2 - 00:00:00 (486 nps)

Query 15: [p4(a82, X0)]

1110 :: 5 - 00:00:02 (479 nps)

Query 16: [p2(X4, X0)]

2 :: 1 - 00:00:00 (64 nps)

Query 17: [p5(a57, a20, X7)]

3 :: 2 - 00:00:00 (618 nps)

Query 18: [p2(X1, X0), p4(X0, a57), p4(X1, a57), p4(X0, a10), p2(a79, X1)]

1893 :: 10 - 00:00:10 (179 nps)

Query 19: [p4(a11, a57)]

4 :: 3 - 00:00:00 (625 nps)

Query 20: [p3(X0, a95), p4(X1, X0), p3(X1, a27), p4(X2, X1), p4(X2, a5)]

71 :: 6 - 00:00:00 (249 nps)

Query 21: [p5(a15, X6, X3)]

2 :: 1 - 00:00:00 (595 nps)

Query 22: [p5(a57, a8, a1)]

3 :: 2 - 00:00:00 (632 nps)

Query 23: [p4(a47, X0)]

4 :: 3 - 00:00:00 (558 nps)

Query 24: [p4(X4, X5)]

2 :: 1 - 00:00:00 (68 nps)

Query 25: [p2(X0, a17), p2(a79, X0)]

6 :: 5 - 00:00:00 (82 nps)

Query 26: [p7(X2, X0), p2(X0, X1), p2(X3, X1), p4(a36, X2), p5(a57, X3, a79)]

9 :: 8 - 00:00:00 (26 nps)

Query 27: [p2(a12, X3)]

3 :: 2 - 00:00:00 (580 nps)

Query 28: [p4(X1, X0), p2(X0, X1), p7(a23, X1), p1(X0, X0), p8(X0, X1, a38, a49)]

351 :: 8 - 00:00:00 (849 nps)

Query 29: [p2(X0, a78), p0(X0, X1), p2(a43, X1), p4(X1, a57)]

21 :: 7 - 00:00:01 (15 nps)

Query 30: [p2(X1, X0), p8(X0, X2, a29, a11), p5(a0, a57, X1), p5(a57, X2, a92)]

10 :: 9 - 00:00:00 (125 nps)

Query 31: [p4(a93, X0), p8(X2, a13, X0, X1), p4(a6, X1), p8(a17, X2, a16, X1)]

7 :: 6 - 00:00:00 (208 nps)

Query 32: [p8(X2, a27, X8, a22)]

3 :: 2 - 00:00:00 (103 nps)

Query 33: [p4(X4, X5)]

2 :: 1 - 00:00:00 (22812 nps)

Query 34: [p5(a57, a10, X0), p5(X2, X0, X1), p4(X0, X2), p5(a11, X0, X1), p2(a25, X2)]

Query failed!!!

230257 :: 0 - 00:00:41 (5534 nps)

Query 35: [p5(X0, a40, X1), p5(X0, a62, a73), p5(X2, a38, X1), p4(a42, X0), p9(X2)]

542 :: 9 - 00:00:03 (157 nps)

Query 36: [p5(X2, X6, a17)]

3 :: 2 - 00:00:00 (533 nps)

Query 37: [p2(X2, X6)]

2 :: 1 - 00:00:00 (22 nps)

Query 38: [p4(a10, X1)]

2 :: 1 - 00:00:00 (165 nps)

Query 39: [p2(X0, a92), p8(X0, a27, a50, a22)]

9 :: 4 - 00:00:00 (65 nps)

Query 40: [p4(X1, X0), p4(a22, X0), p3(X1, a20)]

8 :: 7 - 00:00:00 (30 nps)

Query 41: [p8(a26, a11, X0, a39), p1(X0, X1), p5(a57, X1, a2)]

5 :: 4 - 00:00:00 (72 nps)

Query 42: [p8(X8, X2, a50, X3)]

2 :: 1 - 00:00:00 (881 nps)

Query 43: [p5(X7, X1, a11)]

3 :: 2 - 00:00:00 (483 nps)

Query 44: [p0(X4, X4)]

3 :: 2 - 00:00:00 (424 nps)

Query 45: [p4(a42, X0)]

4 :: 3 - 00:00:00 (37993 nps)

Query 46: [p4(X7, X6)]

2 :: 1 - 00:00:00 (67 nps)

Query 47: [p1(X0, a37), p4(a10, X0)]

4 :: 2 - 00:00:00 (541 nps)

Query 48: [p5(X8, a68, X2)]

4 :: 3 - 00:00:00 (180 nps)

Query 49: [p5(X6, a2, X0)]

3 :: 2 - 00:00:00 (611 nps)

Query 50: [p2(a11, X3)]

3 :: 2 - 00:00:00 (599 nps)

Query 51: [p2(X1, X7)]

2 :: 1 - 00:00:00 (65 nps)

Query 52: [p8(a55, X8, a50, X4)]

927 :: 4 - 00:00:00 (987 nps)

Query 53: [p2(a42, X3)]

3 :: 2 - 00:00:00 (576 nps)

Query 54: [p2(X8, a51)]

3 :: 2 - 00:00:00 (330 nps)

Query 55: [p4(X5, a57)]

5904 :: 3 - 00:00:04 (1299 nps)

Query 56: [p5(a57, a26, X0), p0(a16, X0)]

4 :: 3 - 00:00:00 (75 nps)

Query 57: [p8(X0, a27, X6, X3)]

2 :: 1 - 00:00:00 (203 nps)

Query 58: [p2(a32, a10)]

3 :: 2 - 00:00:00 (204 nps)

Query 59: [p2(X3, a92)]

3 :: 2 - 00:00:00 (66 nps)

Query 60: [p0(a14, X3)]

4 :: 3 - 00:00:00 (109 nps)

Query 61: [p9(X0), p1(a0, X0)]

3 :: 2 - 00:00:00 (80 nps)

Query 62: [p2(X4, X5)]

2 :: 1 - 00:00:00 (63 nps)

Query 63: [p4(X3, X4)]

2 :: 1 - 00:00:00 (36 nps)

Query 64: [p2(X0, X1), p0(X1, X0), p2(X1, a79), p1(a2, X0)]

10112 :: 4 - 00:00:06 (1668 nps)

Query 65: [p4(a13, X5)]

3 :: 2 - 00:00:00 (555 nps)

Query 66: [p2(X1, a14)]

2 :: 1 - 00:00:00 (500 nps)

Query 67: [p2(X0, X1), p0(X1, X0), p5(a57, X1, X2), p1(X1, X2), p3(X1, X2)]
33 :: 6 - 00:00:01 (24 nps)

Query 68: [p3(X0, X0), p5(a57, X0, X0)]
3 :: 2 - 00:00:00 (97 nps)

Query 69: [p8(a30, a27, X1, a22)]
3 :: 2 - 00:00:00 (145 nps)

Query 70: [p5(a57, a50, a0)]
3 :: 2 - 00:00:00 (289 nps)

Query 71: [p8(a85, a18, X0, a43), p8(a5, a16, a42, X0)]
3 :: 2 - 00:00:00 (801 nps)

Query 72: [p2(a4, X0), p4(a82, X0)]
2600 :: 6 - 00:00:06 (383 nps)

Query 73: [p2(X4, a0)]
2 :: 1 - 00:00:00 (109 nps)

Query 74: [p2(X6, X5)]
2 :: 1 - 00:00:00 (28 nps)

Query 75: [p1(a0, X0), p1(X0, X1), p2(X1, X2), p2(a36, X2)]
7 :: 6 - 00:00:00 (16 nps)

Query 76: [p4(X0, a90), p3(X0, X1), p2(X1, a1)]
643 :: 5 - 00:00:05 (119 nps)

Query 77: [p4(X0, a5), p5(a1, X0, a64)]
3 :: 2 - 00:00:00 (395 nps)

Query 78: [p5(X0, X2, X1), p5(X0, a2, X3), p3(X2, X1), p3(X2, a10), p4(X3, a90)]
21 :: 7 - 00:00:00 (50 nps)

Query 79: [p3(X2, a20)]
2 :: 1 - 00:00:00 (473 nps)

Query 80: [p4(X1, X2)]
2 :: 1 - 00:00:00 (67 nps)

Query 81: [p8(X8, a27, a50, X3)]

3 :: 2 - 00:00:00 (100 nps)

Query 82: [p5(a57, X0, a1), p4(a0, X0)]

336 :: 3 - 00:00:00 (811 nps)

Query 83: [p4(X1, X0), p4(a20, X0), p5(X0, X1, a12), p2(a13, X1)]

10 :: 9 - 00:00:00 (147 nps)

Query 84: [p2(X8, a0)]

2 :: 1 - 00:00:00 (276 nps)

Query 85: [p0(X1, X0), p8(X0, a63, a35, a27), p2(a8, X1)]

4 :: 3 - 00:00:00 (287 nps)

Query 86: [p8(a53, X1, a50, X0), p8(X2, X1, a50, X0), p9(X1), p8(X2, a11, a29, a39)]

7 :: 6 - 00:00:00 (62 nps)

Query 87: [p2(X3, X7)]

2 :: 1 - 00:00:00 (63 nps)

Query 88: [p2(X2, a92)]

3 :: 2 - 00:00:00 (137 nps)

Query 89: [p8(a26, a11, a29, X0), p6(a2, X0), p8(a20, a63, X0, a5)]

4 :: 3 - 00:00:00 (560 nps)

Query 90: [p4(X4, X3)]

2 :: 1 - 00:00:00 (66 nps)

Query 91: [p2(X0, a92), p6(X0, a10)]

1161 :: 3 - 00:00:00 (13178 nps)

Query 92: [p5(a87, X0, a12), p1(X0, a1)]

736 :: 2 - 00:00:01 (715 nps)

Query 93: [p4(a84, a52)]

3 :: 2 - 00:00:00 (810 nps)

Query 94: [p8(X2, X0, X1, a22), p8(a10, X0, X1, a22), p8(a92, X0, X1, a22), p2(a1, X2), p2(X1, a92)]

107 :: 10 - 00:00:00 (144 nps)

Query 95: [p5(X0, a57, a23)]
20 :: 4 - 00:00:00 (27 nps)

Query 96: [p2(X0, X1)]
2 :: 1 - 00:00:00 (18966 nps)

Query 97: [p8(X9, a18, a31, X9)]
697 :: 9 - 00:00:00 (1004 nps)

Query 98: [p8(a58, X0, a50, a22), p8(a26, X0, a50, a22)]
5 :: 4 - 00:00:00 (201 nps)

Query 99: [p1(X0, a4), p8(a25, X1, a6, X0), p5(a15, X1, a14)]
4 :: 3 - 00:00:00 (328 nps)

Query 100: [p1(a0, X0), p1(X0, a1)]
5 :: 2 - 00:00:00 (2475 nps)

ming: 2580.22 nodes/query
1 queries failed
Time to run all queries: 92.92171208699993

In []: *#Step 7: change embedding size and do steps 5 and 6 again, make sure to change so that we do not overwrite files from*

In [2]: `%cd size400`

/data/users/jupyter-rem526/cse327-p4-files/size400

In [3]: `!python ../kbencoder.py --kb_path randomKB.txt --train_unification_model --embed_size 25 --embed_model_path randomKB_`

Using cuda device

Uni model embed size: 25

```
[████████████████████] 3500/3500 Anchors generated
[████████████████████] 70000/70000 Triplets generated
[██████-----] 18235/70000 Triplet encodings
```


IOPub data rate exceeded.
 The Jupyter server will temporarily stop sending output
 to the client in order to avoid crashing it.
 To change this limit, set the config variable
 `--ServerApp.iopub\_data\_rate\_limit`.

Current values:
 ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
 ServerApp.rate_limit_window=3.0 (secs)

[██████████]-----] 47048/70000 Triplet encodings

IOPub data rate exceeded.
 The Jupyter server will temporarily stop sending output
 to the client in order to avoid crashing it.
 To change this limit, set the config variable
 `--ServerApp.iopub\_data\_rate\_limit`.

Current values:
 ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
 ServerApp.rate_limit_window=3.0 (secs)

[██████████████████] 70000/70000 Triplet encodings

Early stopping triggered-----] 23.4% Epoch | [Epoch 118/500, Training Loss: 0.2021, Validation Loss: 0.4906]
 Saved model.
 Saved plot.
 unification model generated

In [6]: !python ../nnreasoner.py --embed_type unification -s mr_model_size_25.pt --embed_size 25 --embed_model_path randomKB_

```
States from vocab: vocab
Num predicates: 10
Max arity: 4
Num constants: 100
Num variables: 10
```

```
Training unification
```

```
Embed size: 25
```

```
Reading examples from mr_train_examples.csv
```

```
Parsing examples...
```

```
Generating embeddings...
```

```
Embedding input size: 450
```

```
[██████████] 45017/45017 Prepping examples
```

```
Loaded data...
```

```
Using cuda device
```

```
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
```

```
warnings.warn('Lazy modules are a new feature under heavy development ')
```

```
0      0.17320 (-)
50     0.06292 (-0.0016)
100    0.05855 (-0.0008)
150    0.05706 (-0.0005)
200    0.05545 (-0.0004)
250    0.05471 (-0.0003)
300    0.05382 (-0.0003)
350    0.05315 (-0.0002)
400    0.05234 (-0.0002)
450    0.05154 (-0.0002)
500    0.05095 (-0.0001)
550    0.05037 (-0.0001)
600    0.04992 (-0.0001)
650    0.04949 (-0.0001)
700    0.04910 (-0.0001)
749    0.04839 (-9.2048)
```

```
Saved training loss figure.
```

```
In [40]: !python ../evaluate.py --kb randomKB.txt -u --embed_size 25 --scoring_model_path mr_model_size_25.pt --embed_model_path
```

```
using cuda device
UNITY: ming
    Embedding Model: randomKB_400_e_size_25_model.pth
    Scoring Model: mr_model_size_25.pt

    Control: mingoal
    Rule Eval: Max Rule
Learned Embedding: Network using cuda
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
Query 1: [p4(X5, a14)]
3 :: 2 - 00:00:00 (7 nps)

Query 2: [p4(X3, a57)]
4 :: 3 - 00:00:00 (113 nps)

Query 3: [p9(X0), p3(X0, a12)]
3 :: 2 - 00:00:00 (134 nps)

Query 4: [p2(X0, a28), p4(a21, X0)]
3 :: 2 - 00:00:00 (293 nps)

Query 5: [p7(X0, X1), p7(X0, a1), p6(a2, X1), p6(X1, X2), p2(X2, a99)]
154 :: 5 - 00:00:00 (398 nps)

Query 6: [p4(X3, X0)]
2 :: 1 - 00:00:00 (67 nps)

Query 7: [p4(a1, X4)]
2 :: 1 - 00:00:00 (695 nps)

Query 8: [p8(a40, a27, X5, X0)]
3 :: 2 - 00:00:00 (460 nps)

Query 9: [p2(X0, a99), p7(X0, a16), p6(a39, X0)]
4 :: 3 - 00:00:00 (273 nps)

Query 10: [p2(X1, X7)]
2 :: 1 - 00:00:00 (64 nps)

Query 11: [p5(a82, a12, X0), p7(X1, X0), p6(a36, X1), p1(a36, X1)]
```

5 :: 4 - 00:00:00 (123 nps)

Query 12: [p8(a41, a27, a50, X9)]

3 :: 2 - 00:00:00 (466 nps)

Query 13: [p2(X2, X8)]

2 :: 1 - 00:00:00 (64 nps)

Query 14: [p8(a71, a27, X5, a22)]

3 :: 2 - 00:00:00 (469 nps)

Query 15: [p4(a82, X0)]

5167 :: 5 - 00:00:05 (928 nps)

Query 16: [p2(X4, X0)]

2 :: 1 - 00:00:00 (64 nps)

Query 17: [p5(a57, a20, X7)]

3 :: 2 - 00:00:00 (587 nps)

Query 18: [p2(X1, X0), p4(X0, a57), p4(X1, a57), p4(X0, a10), p2(a79, X1)]

883 :: 10 - 00:00:03 (232 nps)

Query 19: [p4(a11, a57)]

4 :: 3 - 00:00:00 (641 nps)

Query 20: [p3(X0, a95), p4(X1, X0), p3(X1, a27), p4(X2, X1), p4(X2, a5)]

160 :: 6 - 00:00:02 (78 nps)

Query 21: [p5(a15, X6, X3)]

2 :: 1 - 00:00:00 (589 nps)

Query 22: [p5(a57, a8, a1)]

3 :: 2 - 00:00:00 (616 nps)

Query 23: [p4(a47, X0)]

4 :: 3 - 00:00:00 (541 nps)

Query 24: [p4(X4, X5)]

2 :: 1 - 00:00:00 (66 nps)

Query 25: [p2(X0, a17), p2(a79, X0)]

6 :: 5 - 00:00:00 (67 nps)

Query 26: [p7(X2, X0), p2(X0, X1), p2(X3, X1), p4(a36, X2), p5(a57, X3, a79)]
1545 :: 8 - 00:00:03 (501 nps)

Query 27: [p2(a12, X3)]
3 :: 2 - 00:00:00 (572 nps)

Query 28: [p4(X1, X0), p2(X0, X1), p7(a23, X1), p1(X0, X0), p8(X0, X1, a38, a49)]
9 :: 8 - 00:00:00 (172 nps)

Query 29: [p2(X0, a78), p0(X0, X1), p2(a43, X1), p4(X1, a57)]
21468 :: 7 - 00:00:08 (2501 nps)

Query 30: [p2(X1, X0), p8(X0, X2, a29, a11), p5(a0, a57, X1), p5(a57, X2, a92)]
10 :: 9 - 00:00:00 (107 nps)

Query 31: [p4(a93, X0), p8(X2, a13, X0, X1), p4(a6, X1), p8(a17, X2, a16, X1)]
7 :: 6 - 00:00:00 (207 nps)

Query 32: [p8(X2, a27, X8, a22)]
3 :: 2 - 00:00:00 (53 nps)

Query 33: [p4(X4, X5)]
2 :: 1 - 00:00:00 (22467 nps)

Query 34: [p5(a57, a10, X0), p5(X2, X0, X1), p4(X0, X2), p5(a11, X0, X1), p2(a25, X2)]
Query failed!!!
39593 :: 0 - 00:00:22 (1725 nps)

Query 35: [p5(X0, a40, X1), p5(X0, a62, a73), p5(X2, a38, X1), p4(a42, X0), p9(X2)]
10 :: 9 - 00:00:00 (56 nps)

Query 36: [p5(X2, X6, a17)]
3 :: 2 - 00:00:00 (481 nps)

Query 37: [p2(X2, X6)]
2 :: 1 - 00:00:00 (62 nps)

Query 38: [p4(a10, X1)]
2 :: 1 - 00:00:00 (446 nps)

Query 39: [p2(X0, a92), p8(X0, a27, a50, a22)]

5 :: 4 - 00:00:00 (67 nps)

Query 40: [p4(X1, X0), p4(a22, X0), p3(X1, a20)]

8 :: 7 - 00:00:00 (57 nps)

Query 41: [p8(a26, a11, X0, a39), p1(X0, X1), p5(a57, X1, a2)]

6 :: 4 - 00:00:00 (83 nps)

Query 42: [p8(X8, X2, a50, X3)]

2 :: 1 - 00:00:00 (789 nps)

Query 43: [p5(X7, X1, a11)]

3 :: 2 - 00:00:00 (472 nps)

Query 44: [p0(X4, X4)]

2759 :: 2 - 00:00:01 (1554 nps)

Query 45: [p4(a42, X0)]

4 :: 3 - 00:00:00 (695 nps)

Query 46: [p4(X7, X6)]

2 :: 1 - 00:00:00 (65 nps)

Query 47: [p1(X0, a37), p4(a10, X0)]

5 :: 2 - 00:00:00 (620 nps)

Query 48: [p5(X8, a68, X2)]

4 :: 3 - 00:00:00 (175 nps)

Query 49: [p5(X6, a2, X0)]

3 :: 2 - 00:00:00 (578 nps)

Query 50: [p2(a11, X3)]

3 :: 2 - 00:00:00 (583 nps)

Query 51: [p2(X1, X7)]

2 :: 1 - 00:00:00 (61 nps)

Query 52: [p8(a55, X8, a50, X4)]

1244 :: 4 - 00:00:00 (5964 nps)

Query 53: [p2(a42, X3)]

3 :: 2 - 00:00:00 (566 nps)

Query 54: [p2(X8, a51)]

3 :: 2 - 00:00:00 (326 nps)

Query 55: [p4(X5, a57)]

4 :: 3 - 00:00:00 (114 nps)

Query 56: [p5(a57, a26, X0), p0(a16, X0)]

4 :: 3 - 00:00:00 (232 nps)

Query 57: [p8(X0, a27, X6, X3)]

2 :: 1 - 00:00:00 (370 nps)

Query 58: [p2(a32, a10)]

3 :: 2 - 00:00:00 (543 nps)

Query 59: [p2(X3, a92)]

3 :: 2 - 00:00:00 (138 nps)

Query 60: [p0(a14, X3)]

161 :: 3 - 00:00:00 (994 nps)

Query 61: [p9(X0), p1(a0, X0)]

14 :: 2 - 00:00:00 (519 nps)

Query 62: [p2(X4, X5)]

2 :: 1 - 00:00:00 (63 nps)

Query 63: [p4(X3, X4)]

2 :: 1 - 00:00:00 (66 nps)

Query 64: [p2(X0, X1), p0(X1, X0), p2(X1, a79), p1(a2, X0)]

32907 :: 5 - 00:00:09 (3561 nps)

Query 65: [p4(a13, X5)]

3 :: 2 - 00:00:00 (531 nps)

Query 66: [p2(X1, a14)]

2 :: 1 - 00:00:00 (472 nps)

Query 67: [p2(X0, X1), p0(X1, X0), p5(a57, X1, X2), p1(X1, X2), p3(X1, X2)]
78293 :: 5 - 00:00:24 (3237 nps)

Query 68: [p3(X0, X0), p5(a57, X0, X0)]
3 :: 2 - 00:00:00 (282 nps)

Query 69: [p8(a30, a27, X1, a22)]
3 :: 2 - 00:00:00 (423 nps)

Query 70: [p5(a57, a50, a0)]
3 :: 2 - 00:00:00 (574 nps)

Query 71: [p8(a85, a18, X0, a43), p8(a5, a16, a42, X0)]
3 :: 2 - 00:00:00 (786 nps)

Query 72: [p2(a4, X0), p4(a82, X0)]
6295 :: 6 - 00:00:04 (1492 nps)

Query 73: [p2(X4, a0)]
2 :: 1 - 00:00:00 (278 nps)

Query 74: [p2(X6, X5)]
2 :: 1 - 00:00:00 (64 nps)

Query 75: [p1(a0, X0), p1(X0, X1), p2(X1, X2), p2(a36, X2)]
82 :: 6 - 00:00:00 (96 nps)

Query 76: [p4(X0, a90), p3(X0, X1), p2(X1, a1)]
7004 :: 5 - 00:00:06 (1070 nps)

Query 77: [p4(X0, a5), p5(a1, X0, a64)]
207 :: 2 - 00:00:00 (782 nps)

Query 78: [p5(X0, X2, X1), p5(X0, a2, X3), p3(X2, X1), p3(X2, a10), p4(X3, a90)]
8 :: 7 - 00:00:00 (29 nps)

Query 79: [p3(X2, a20)]
2 :: 1 - 00:00:00 (461 nps)

Query 80: [p4(X1, X2)]
2 :: 1 - 00:00:00 (66 nps)

Query 81: [p8(X8, a27, a50, X3)]

3 :: 2 - 00:00:00 (101 nps)

Query 82: [p5(a57, X0, a1), p4(a0, X0)]

60 :: 3 - 00:00:01 (38 nps)

Query 83: [p4(X1, X0), p4(a20, X0), p5(X0, X1, a12), p2(a13, X1)]

360 :: 9 - 00:00:00 (380 nps)

Query 84: [p2(X8, a0)]

2 :: 1 - 00:00:00 (231 nps)

Query 85: [p0(X1, X0), p8(X0, a63, a35, a27), p2(a8, X1)]

4 :: 3 - 00:00:00 (77 nps)

Query 86: [p8(a53, X1, a50, X0), p8(X2, X1, a50, X0), p9(X1), p8(X2, a11, a29, a39)]

7 :: 6 - 00:00:00 (50 nps)

Query 87: [p2(X3, X7)]

2 :: 1 - 00:00:00 (55 nps)

Query 88: [p2(X2, a92)]

3 :: 2 - 00:00:00 (117 nps)

Query 89: [p8(a26, a11, a29, X0), p6(a2, X0), p8(a20, a63, X0, a5)]

4 :: 3 - 00:00:00 (482 nps)

Query 90: [p4(X4, X3)]

2 :: 1 - 00:00:00 (59 nps)

Query 91: [p2(X0, a92), p6(X0, a10)]

5 :: 3 - 00:00:00 (165 nps)

Query 92: [p5(a87, X0, a12), p1(X0, a1)]

3 :: 2 - 00:00:00 (388 nps)

Query 93: [p4(a84, a52)]

3 :: 2 - 00:00:00 (764 nps)

Query 94: [p8(X2, X0, X1, a22), p8(a10, X0, X1, a22), p8(a92, X0, X1, a22), p2(a1, X2), p2(X1, a92)]

144 :: 13 - 00:00:00 (280 nps)

Query 95: [p5(X0, a57, a23)]
20 :: 4 - 00:00:00 (23 nps)

Query 96: [p2(X0, X1)]
2 :: 1 - 00:00:00 (65 nps)

Query 97: [p8(X9, a18, a31, X9)]
4 :: 3 - 00:00:00 (171 nps)

Query 98: [p8(a58, X0, a50, a22), p8(a26, X0, a50, a22)]
5 :: 4 - 00:00:00 (214 nps)

Query 99: [p1(X0, a4), p8(a25, X1, a6, X0), p5(a15, X1, a14)]
4 :: 3 - 00:00:00 (301 nps)

Query 100: [p1(a0, X0), p1(X0, a1)]
4 :: 2 - 00:00:00 (8741 nps)

ming: 1987.99 nodes/query
1 queries failed
Time to run all queries: 100.41778309299997

In []: *#Step 8: use modified nnreasoner to reperform on 400 random KB, make sure to rename files, and pull correct nnreasoner
for simplicity, I replaced the code in nnreasoner.py with that in nnreasoner-modified.py, to avoid having to change*

In [9]: `!python ../kbencoder.py --kb_path randomKB.txt --train_unification_model --embed_model_path randomKB_400_modified_rea`

```
Using cuda device
Uni model embed size: 50
[████████████████████] 3500/3500 Anchors generated
[████████████████████] 70000/70000 Triplets generated
[████████-----] 18241/70000 Triplet encodings
```

IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
ServerApp.rate_limit_window=3.0 (secs)

```
[██████████]-----] 46887/70000 Triplet encodings
```

IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
[██████████] 70000/70000 Triplet encodings
```

Early stopping triggered-----] 25.7% Epoch | [Epoch 129/500, Training Loss: 0.1662, Validation Loss: 0.4546]

Saved model.

Saved plot.

unification model generated

```
In [62]: !python ../nnreasoner.py --embed_type unification -s mr_model_modified_reasoner.pt --embed_model_path randomKB_400_mc
```

States from vocab: vocab
Num predicates: 10
Max arity: 4
Num constants: 100
Num variables: 10

Training unification

Embed size: 50

Reading examples from mr_train_examples.csv

Parsing examples...

Generating embeddings...

Embedding input size: 450

[██████████] 45017/45017 Prepping examples

Loaded data...

Using cuda device

/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.

warnings.warn('Lazy modules are a new feature under heavy development ')

0	0.55318 (-)
50	0.05777 (-0.0065)
100	0.05468 (-0.0029)
150	0.05299 (-0.0020)
200	0.05098 (-0.0015)
250	0.05015 (-0.0012)
300	0.04839 (-0.0010)
350	0.04725 (-0.0009)
400	0.04640 (-0.0007)
450	0.04569 (-0.0007)
500	0.04484 (-0.0006)
550	0.04466 (-0.0005)
600	0.04389 (-0.0005)
650	0.04325 (-0.0004)
700	0.04278 (-0.0004)
750	0.04194 (-0.0004)
800	0.04194 (-0.0004)
850	0.04136 (-0.0003)
900	0.04150 (-0.0003)
950	0.04084 (-0.0003)
1000	0.04025 (-0.0003)
1050	0.03978 (-0.0003)
1100	0.03987 (-0.0002)
1150	0.03956 (-0.0002)

```
1200    0.03933 (-0.0002)
1250    0.03889 (-0.0002)
1300    0.03907 (-0.0002)
1350    0.03845 (-0.0002)
1400    0.03837 (-0.0002)
1449    0.03804 (-0.0001)
Saved training loss figure.
```

```
In [63]: !python ../evaluate.py --kb randomKB.txt -u --embed_model_path randomKB_400_modified_reasoner.pth --scoring_model_pat
```

```

using cuda device
UNITY: ming
    Embedding Model: randomKB_400_modified_reasoner.pth
    Scoring Model: mr_model_modified_reasoner.pt

    Control: mingoal
    Rule Eval: Max Rule
Learned Embedding: Network using cuda
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
Query 1: [p4(X5, a14)]
3 :: 2 - 00:00:00 (6 nps)

Query 2: [p4(X3, a57)]
4 :: 3 - 00:00:00 (36 nps)

Query 3: [p9(X0), p3(X0, a12)]
3 :: 2 - 00:00:00 (48 nps)

Query 4: [p2(X0, a28), p4(a21, X0)]
3 :: 2 - 00:00:00 (94 nps)

Query 5: [p7(X0, X1), p7(X0, a1), p6(a2, X1), p6(X1, X2), p2(X2, a99)]
631 :: 5 - 00:00:05 (118 nps)

Query 6: [p4(X3, X0)]
2 :: 1 - 00:00:00 (22 nps)

Query 7: [p4(a1, X4)]
2 :: 1 - 00:00:00 (244 nps)

Query 8: [p8(a40, a27, X5, X0)]
3 :: 2 - 00:00:00 (421 nps)

Query 9: [p2(X0, a99), p7(X0, a16), p6(a39, X0)]
841 :: 3 - 00:00:01 (792 nps)

Query 10: [p2(X1, X7)]
2 :: 1 - 00:00:00 (25 nps)

Query 11: [p5(a82, a12, X0), p7(X1, X0), p6(a36, X1), p1(a36, X1)]

```

5 :: 4 - 00:00:00 (42 nps)

Query 12: [p8(a41, a27, a50, X9)]

3 :: 2 - 00:00:00 (153 nps)

Query 13: [p2(X2, X8)]

2 :: 1 - 00:00:00 (20 nps)

Query 14: [p8(a71, a27, X5, a22)]

3 :: 2 - 00:00:00 (153 nps)

Query 15: [p4(a82, X0)]

20 :: 5 - 00:00:02 (8 nps)

Query 16: [p2(X4, X0)]

2 :: 1 - 00:00:00 (26 nps)

Query 17: [p5(a57, a20, X7)]

3 :: 2 - 00:00:00 (236 nps)

Query 18: [p2(X1, X0), p4(X0, a57), p4(X1, a57), p4(X0, a10), p2(a79, X1)]

739 :: 10 - 00:00:06 (110 nps)

Query 19: [p4(a11, a57)]

4 :: 3 - 00:00:00 (214 nps)

Query 20: [p3(X0, a95), p4(X1, X0), p3(X1, a27), p4(X2, X1), p4(X2, a5)]

70 :: 6 - 00:00:01 (38 nps)

Query 21: [p5(a15, X6, X3)]

2 :: 1 - 00:00:00 (135 nps)

Query 22: [p5(a57, a8, a1)]

3 :: 2 - 00:00:00 (151 nps)

Query 23: [p4(a47, X0)]

4 :: 3 - 00:00:00 (175 nps)

Query 24: [p4(X4, X5)]

2 :: 1 - 00:00:00 (20 nps)

Query 25: [p2(X0, a17), p2(a79, X0)]

6 :: 5 - 00:00:00 (32 nps)

Query 26: [p7(X2, X0), p2(X0, X1), p2(X3, X1), p4(a36, X2), p5(a57, X3, a79)]

9 :: 8 - 00:00:00 (11 nps)

Query 27: [p2(a12, X3)]

1546 :: 2 - 00:00:06 (228 nps)

Query 28: [p4(X1, X0), p2(X0, X1), p7(a23, X1), p1(X0, X0), p8(X0, X1, a38, a49)]

9 :: 8 - 00:00:00 (152 nps)

Query 29: [p2(X0, a78), p0(X0, X1), p2(a43, X1), p4(X1, a57)]

21 :: 7 - 00:00:01 (14 nps)

Query 30: [p2(X1, X0), p8(X0, X2, a29, a11), p5(a0, a57, X1), p5(a57, X2, a92)]

10 :: 9 - 00:00:00 (107 nps)

Query 31: [p4(a93, X0), p8(X2, a13, X0, X1), p4(a6, X1), p8(a17, X2, a16, X1)]

7 :: 6 - 00:00:00 (282 nps)

Query 32: [p8(X2, a27, X8, a22)]

3 :: 2 - 00:00:00 (90 nps)

Query 33: [p4(X4, X5)]

2 :: 1 - 00:00:00 (21771 nps)

Query 34: [p5(a57, a10, X0), p5(X2, X0, X1), p4(X0, X2), p5(a11, X0, X1), p2(a25, X2)]

Query failed!!!

83330 :: 0 - 00:00:44 (1879 nps)

Query 35: [p5(X0, a40, X1), p5(X0, a62, a73), p5(X2, a38, X1), p4(a42, X0), p9(X2)]

144 :: 9 - 00:00:01 (127 nps)

Query 36: [p5(X2, X6, a17)]

3 :: 2 - 00:00:00 (170 nps)

Query 37: [p2(X2, X6)]

2 :: 1 - 00:00:00 (22 nps)

Query 38: [p4(a10, X1)]

2 :: 1 - 00:00:00 (119 nps)

Query 39: [p2(X0, a92), p8(X0, a27, a50, a22)]

6 :: 4 - 00:00:00 (24 nps)

Query 40: [p4(X1, X0), p4(a22, X0), p3(X1, a20)]

8 :: 7 - 00:00:00 (15 nps)

Query 41: [p8(a26, a11, X0, a39), p1(X0, X1), p5(a57, X1, a2)]

5 :: 4 - 00:00:00 (62 nps)

Query 42: [p8(X8, X2, a50, X3)]

2 :: 1 - 00:00:00 (790 nps)

Query 43: [p5(X7, X1, a11)]

3 :: 2 - 00:00:00 (430 nps)

Query 44: [p0(X4, X4)]

3 :: 2 - 00:00:00 (382 nps)

Query 45: [p4(a42, X0)]

4 :: 3 - 00:00:00 (628 nps)

Query 46: [p4(X7, X6)]

2 :: 1 - 00:00:00 (17 nps)

Query 47: [p1(X0, a37), p4(a10, X0)]

3 :: 2 - 00:00:00 (147 nps)

Query 48: [p5(X8, a68, X2)]

4 :: 3 - 00:00:00 (110 nps)

Query 49: [p5(X6, a2, X0)]

3 :: 2 - 00:00:00 (196 nps)

Query 50: [p2(a11, X3)]

3 :: 2 - 00:00:00 (202 nps)

Query 51: [p2(X1, X7)]

2 :: 1 - 00:00:00 (42 nps)

Query 52: [p8(a55, X8, a50, X4)]

80 :: 4 - 00:00:00 (419 nps)

Query 53: [p2(a42, X3)]
1243 :: 2 - 00:00:07 (162 nps)

Query 54: [p2(X8, a51)]
3 :: 2 - 00:00:00 (82 nps)

Query 55: [p4(X5, a57)]
4 :: 3 - 00:00:00 (30 nps)

Query 56: [p5(a57, a26, X0), p0(a16, X0)]
4 :: 3 - 00:00:00 (56 nps)

Query 57: [p8(X0, a27, X6, X3)]
2 :: 1 - 00:00:00 (152 nps)

Query 58: [p2(a32, a10)]
3 :: 2 - 00:00:00 (167 nps)

Query 59: [p2(X3, a92)]
3 :: 2 - 00:00:00 (44 nps)

Query 60: [p0(a14, X3)]
4 :: 3 - 00:00:00 (73 nps)

Query 61: [p9(X0), p1(a0, X0)]
3 :: 2 - 00:00:00 (40 nps)

Query 62: [p2(X4, X5)]
2 :: 1 - 00:00:00 (18 nps)

Query 63: [p4(X3, X4)]
2 :: 1 - 00:00:00 (17 nps)

Query 64: [p2(X0, X1), p0(X1, X0), p2(X1, a79), p1(a2, X0)]
790 :: 4 - 00:00:00 (839 nps)

Query 65: [p4(a13, X5)]
3 :: 2 - 00:00:00 (136 nps)

Query 66: [p2(X1, a14)]
2 :: 1 - 00:00:00 (119 nps)

Query 67: [p2(X0, X1), p0(X1, X0), p5(a57, X1, X2), p1(X1, X2), p3(X1, X2)]

497 :: 6 - 00:00:25 (19 nps)

Query 68: [p3(X0, X0), p5(a57, X0, X0)]

3 :: 2 - 00:00:00 (133 nps)

Query 69: [p8(a30, a27, X1, a22)]

3 :: 2 - 00:00:00 (230 nps)

Query 70: [p5(a57, a50, a0)]

3 :: 2 - 00:00:00 (244 nps)

Query 71: [p8(a85, a18, X0, a43), p8(a5, a16, a42, X0)]

3 :: 2 - 00:00:00 (499 nps)

Query 72: [p2(a4, X0), p4(a82, X0)]

21 :: 6 - 00:00:01 (12 nps)

Query 73: [p2(X4, a0)]

2 :: 1 - 00:00:00 (149 nps)

Query 74: [p2(X6, X5)]

2 :: 1 - 00:00:00 (25 nps)

Query 75: [p1(a0, X0), p1(X0, X1), p2(X1, X2), p2(a36, X2)]

34619 :: 6 - 00:00:22 (1523 nps)

Query 76: [p4(X0, a90), p3(X0, X1), p2(X1, a1)]

1554 :: 5 - 00:00:03 (389 nps)

Query 77: [p4(X0, a5), p5(a1, X0, a64)]

3 :: 2 - 00:00:00 (195 nps)

Query 78: [p5(X0, X2, X1), p5(X0, a2, X3), p3(X2, X1), p3(X2, a10), p4(X3, a90)]

8 :: 7 - 00:00:00 (17 nps)

Query 79: [p3(X2, a20)]

2 :: 1 - 00:00:00 (229 nps)

Query 80: [p4(X1, X2)]

2 :: 1 - 00:00:00 (25 nps)

Query 81: [p8(X8, a27, a50, X3)]

3 :: 2 - 00:00:00 (27 nps)

Query 82: [p5(a57, X0, a1), p4(a0, X0)]

18 :: 3 - 00:00:01 (16 nps)

Query 83: [p4(X1, X0), p4(a20, X0), p5(X0, X1, a12), p2(a13, X1)]

137 :: 9 - 00:00:01 (75 nps)

Query 84: [p2(X8, a0)]

2 :: 1 - 00:00:00 (99 nps)

Query 85: [p0(X1, X0), p8(X0, a63, a35, a27), p2(a8, X1)]

4 :: 3 - 00:00:00 (27 nps)

Query 86: [p8(a53, X1, a50, X0), p8(X2, X1, a50, X0), p9(X1), p8(X2, a11, a29, a39)]

7 :: 6 - 00:00:00 (41 nps)

Query 87: [p2(X3, X7)]

2 :: 1 - 00:00:00 (21 nps)

Query 88: [p2(X2, a92)]

3 :: 2 - 00:00:00 (55 nps)

Query 89: [p8(a26, a11, a29, X0), p6(a2, X0), p8(a20, a63, X0, a5)]

4 :: 3 - 00:00:00 (209 nps)

Query 90: [p4(X4, X3)]

2 :: 1 - 00:00:00 (19 nps)

Query 91: [p2(X0, a92), p6(X0, a10)]

5 :: 3 - 00:00:00 (57 nps)

Query 92: [p5(a87, X0, a12), p1(X0, a1)]

766 :: 2 - 00:00:00 (774 nps)

Query 93: [p4(a84, a52)]

3 :: 2 - 00:00:00 (756 nps)

Query 94: [p8(X2, X0, X1, a22), p8(a10, X0, X1, a22), p8(a92, X0, X1, a22), p2(a1, X2), p2(X1, a92)]

11 :: 10 - 00:00:00 (14 nps)

Query 95: [p5(X0, a57, a23)]
20 :: 4 - 00:00:01 (19 nps)

Query 96: [p2(X0, X1)]
2 :: 1 - 00:00:00 (22 nps)

Query 97: [p8(X9, a18, a31, X9)]
4 :: 3 - 00:00:00 (82 nps)

Query 98: [p8(a58, X0, a50, a22), p8(a26, X0, a50, a22)]
5 :: 4 - 00:00:00 (94 nps)

Query 99: [p1(X0, a4), p8(a25, X1, a6, X0), p5(a15, X1, a14)]
4 :: 3 - 00:00:00 (161 nps)

Query 100: [p1(a0, X0), p1(X0, a1)]
6 :: 2 - 00:00:00 (421 nps)

ming: 1273.76 nodes/query

1 queries failed

Time to run all queries: 145.43377977799997

In []: